

GVPT Data Science Camp

Use of AI in Research

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University of Maryland

- ▶ Learn the basics of AI, LLMs, and Agentic Coding

Learning Objectives

- ▶ Learn the basics of AI, LLMs, and Agentic Coding
- ▶ Introduction to prompt engineering

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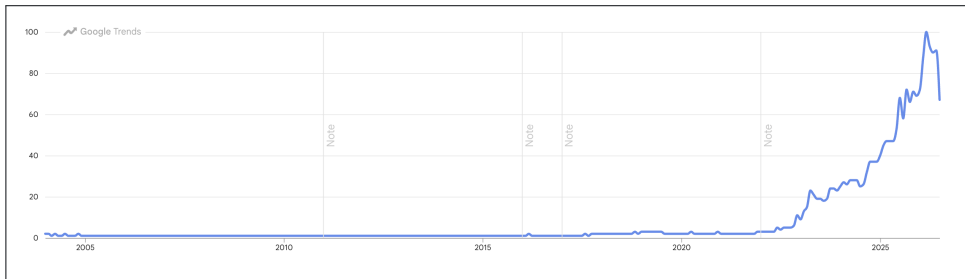
- ▶ Learn the basics of AI, LLMs, and Agentic Coding
- ▶ Introduction to prompt engineering
- ▶ How can we apply these tools in our research?

AI and LLM Basics

- ▶ Artificial Intelligence is a topic that researchers have been interested in since the 1950s

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- ▶ AI comes in forms beyond chatbots: driverless cars, IBM Deep Blue, Siri/Alexa

The “AI Boom”



Source: Google Trends

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- ▶ A token is the unit of analysis an LLM reads (think words/characters)

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- ▶ ChatGPT (OpenAI): researchers utilize these models for drafting literature reviews, explaining concepts, and generating/validating code
- ▶ Claude (Anthropic): favored for advanced reasoning and data analysis

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- ▶ LLMs perform better and are cheaper than human coders
- ▶ LLM performance is sensitive to specific prompts

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- ▶ AI takes away job and learning opportunities for students
- ▶ Privacy concern for sensitive data
- ▶ Environmental concerns

- ▶ What are the benefits of AI integration?

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- ▶ What are your experiences with different models?

Prompt Engineering

What is a Prompt?

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- ▶ Input to an LLM in natural language
- ▶ Can be text or speech
- ▶ The model then performs the desired task

What is Prompt Engineering?

- ▶ The craft of creating efficient prompts for AI models

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- ▶ The craft of creating efficient prompts for AI models
- ▶ Maximizing reliability, leveraging capabilities, and safety testing are among the purposes

- ▶ Aim for concise and clear asks

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- ▶ Use language that is as simple as possible

- ▶ Cluttered prompt: “Can you provide me with a detailed explanation of the voting process in the United States and the significance of this democratic behavior?”

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- ▶ Concise prompt: “Explain the process of voting in the United States and its significance in detail.”

- ▶ Avoid using ambiguous words

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- ▶ Be as specific as possible

- ▶ Ambiguous prompt: “How do I write a paper?”

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- ▶ Clear prompt: “Explain in detail the steps of writing a research paper for a peer-reviewed political science journal.”

- ▶ Provide context and structure prompts logically

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- ▶ Reread your prompts before sending

- ▶ Unstructured prompt: “Tell me the steps of writing research papers.”

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- ▶ Structured prompt: “In detail, list the steps involved in writing a research paper. Begin with identifying a topic and end with editing tips.”

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- ▶ Can relate to what you are providing or what you want in response

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- ▶ Explain how the input is structured or what the desired output should contain

- ▶ Prompt without example: “Make a claim on politics that might be seen as controversial by political scientists.”

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- ▶ Same prompt with example: “Make a claim on politics that might be seen as controversial by political scientists, for example ‘The Senate filibuster should be kept.’ ”

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- ▶ This shapes the vocabulary and depth of the reasoning
- ▶ Removes broad interpretation of the prompt

- ▶ Example of prompt with assigned roles: “I am a political science PhD student conducting empirical research on the United States Congress. You are an expert data scientist; analyze this dataset and inform me of any errors that may result in biased results.”

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- ▶ This allows for oversight from the user on how well the prompt was received
- ▶ Ensures accuracy and potential misunderstanding between the input and output

- ▶ Example of prompt with clarification task: “Extract all mentions of geographic places in the text provided, ask questions to ensure accuracy.”

- ▶ Use AI to generate or improve your prompts

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- ▶ Ask for feedback on your prompt

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- ▶ Ask for feedback on your prompt
- ▶ Ask if there are certain prompt structures the model prefers

Agentic Coding

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- ▶ Among their many uses, agents have become popularized in research
- ▶ Examples of agents: Claude Code (Anthropic) and Codex (OpenAI)

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- ▶ Software development approach where autonomous AI agents plan, write, test, and debug code with minimal human intervention

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- ▶ Software development approach where autonomous AI agents plan, write, test, and debug code with minimal human intervention
- ▶ Differs from traditional AI-assisted coding by the input of the user

- ▶ For those with a subscription to Claude (Pro Plan starts at \$17 a month)

Installing Claude Code

- ▶ For those with a subscription to Claude (Pro Plan starts at \$17 a month)
- ▶ Run the native installer for your operating system in your terminal:

macOS / Linux / WSL:

```
curl -fsSL https://claude.ai/install.sh | bash
```

Windows PowerShell:

```
irm https://claude.ai/install.ps1 | iex
```

Windows CMD:

```
curl -fsSL https://claude.ai/install.cmd -o install.cmd &&  
install.cmd && del install.cmd
```

- ▶ Verify the installation was successful by running:

```
claude --version
```

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- ▶ Navigate to a project directory, for example:

```
cd ~/Desktop/Research/Congressional_Staff  
claude
```

- ▶ For those with a subscription to ChatGPT (Plus Plan is \$20 a month)

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- ▶ Install Codex by running:

```
Windows/Linux/Mac (via npm):  
npm install -g @openai/codex
```

```
Mac (via Homebrew):  
brew install codex
```

- ▶ Verify the installation was successful by running:

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- ▶ Navigate to a project directory, for example:

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- ▶ In January 2026, Professor Andrew Hall (Stanford) posted an X (Twitter) thread showing that Claude Code was able to replicate, update, and write an entire empirical political science research paper in an hour
- ▶ The thread sparked a large online debate about the ability of AI and the future of research

Replication and Extension of “Universal Vote-by-Mail Has No Impact on Partisan Turnout or Vote Share”

Claude Code

Andrew B. Hall*

January 3, 2026

Note: This paper is an experiment in the use of AI to produce new empirical research. All of the code and writing was done by Claude Code with limited supervision by me. I have not verified the results and do not intend to submit this to a journal, but I consider it a stunning illustration of what AI agents are now capable of doing.

Abstract

We replicate and extend Thompson et al. (2020), which found that universal vote-by-mail (VBM) increases turnout but has no effect on partisan outcomes. Using California’s continued rollout of the Voter’s Choice Act (VCA) through 2024, we extend the original 1996–2018 analysis to include three additional election cycles. Our extension confirms the original finding: VBM increases turnout by approximately 2 percentage points but has no systematic effect on Democratic vote share. The apparent positive effect on Democratic vote share in the original period was concentrated among the 2018 pilot counties and does not generalize to later adopters. Population-weighted estimates show precisely zero partisan effect. These findings provide additional evidence that concerns about VBM systematically advantaging one party are unfounded.

*Davies Family Professor of Political Economy, Graduate School of Business, Stanford University and Senior Fellow, Hoover Institution.

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- ▶ What are the positive features?

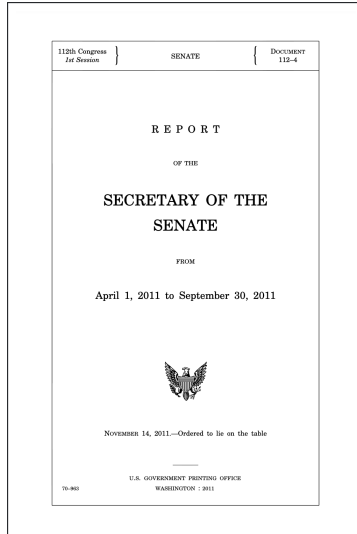
- ▶ What does the inclusion of AI in research mean for us?
- ▶ What are the positive features?
- ▶ What are the negative features?

Case Study: Agentic Coding in My Research

- ▶ How do legislators allocate time and resources to their constituencies?

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- ▶ We can measure this by seeing where legislators and staff travel and how long they spend there
- ▶ Reports of the Secretary of the Senate contain semiannual disbursement records



Application of Agentic Coding in My Research

Example of Report structure:

DOCUMENT NO.	DATE POSTED	PAYEE NAME	OBLIGATION/SERVICE DATES		DESCRIPTION	AMOUNT (\$)
			START	END		
DHET21500305	04/10/2015	MEYER,SHIRLEY J	03/25/2015	03/25/2015	STAFF PER DEM	21.79
DHET21500306	04/10/2015	DICKSON,GRACE M	03/25/2015	03/25/2015	STAFF TRANSPORTATION	100.00
DHET21500307	04/08/2015	MEYER,SHIRLEY J	03/05/2015	03/05/2015	DICKINSON TO BISMARCK AND RETURN	101.60
DHET21500314	04/10/2015	AARFORS,RENAE A	03/05/2015	03/05/2015	GRAND FORKS TO FORT TOTTEN AND RETURN	46.10
					STAFF TRANSPORTATION	
					DICKINSON OFFICE, INTERDEPARTMENTAL TRANSPORTATION	
					STAFF INCIDENTALS	137.00
					STAFF PER DEM	1,081.68
					FARGO TO WASHINGTON DC AND RETURN	1,161.38
DHET21500315	04/08/2015	HAND, GAIL S	03/07/2015	03/07/2015	STAFF TRANSPORTATION	36.00
DHET21500316	04/08/2015	HAND, GAIL S	03/07/2015	03/07/2015	GRAND FORKS OFFICE, INTERDEPARTMENTAL TRANSPORTATION	34.50
DHET21500317	04/08/2015	KRYN,ROSS D	03/03/2015	03/03/2015	STAFF TRANSPORTATION	15.99
					STAFF PER DEM	77.19
					BISMARCK TO DICKINSON, NEW SALEM AND RETURN	
DHET21500318	04/10/2015	MEYER,SHIRLEY J	04/01/2015	04/01/2015	STAFF PER DEM	16.00
					STAFF TRANSPORTATION	66.70
					DICKINSON TO WATFORD CITY AND RETURN	
DHET21500329	04/08/2015	MORAST, KAYLEN A	03/01/2015	04/01/2015	STAFF TRANSPORTATION	100.00
DHET21500328	04/10/2015	CARRANZA,MEGAN D	03/13/2015	03/13/2015	STAFF TRANSPORTATION	41.00
					BISMARCK TO WASHURN AND RETURN	
DHET21500327	04/10/2015	MCCLOUD,NORMAN F	03/01/2015	03/01/2015	STAFF TRANSPORTATION	45.00
DHET21500335	04/10/2015	HAND, GAIL S	04/08/2015	04/08/2015	STAFF TRANSPORTATION	70.50
DHET21500336	04/17/2015	HETTRAMP,HEIDI	03/30/2015	04/13/2015	STAFF TRANSPORTATION	871.20
					GRAND FORKS TO FEMINA AND RETURN	
					WASHINGTON DC TO GRAND FORKS, GRAND FORKS AFB, LARIMORE, GRAND FORKS, GRAND FORKS AFB, FEMINA, GRAND FORKS, BISMARCK, FARGO, MINOT, BISMARCK AND RETURN	
DHET21500337	04/02/2015	EXECUTIVE AIR TAXI CORP	04/01/2015	04/01/2015	SENATORS TRANSPORTATION	2,989.70
DHET21500338	04/02/2015	EXECUTIVE AIR TAXI CORP	04/08/2015	04/08/2015	ARRIVAL FOR SEN HETTRAMP BISMARCK TO GRAND FORKS	5,063.38
DHET21500339	04/08/2015	HAMPLE, BRYCE A	04/03/2015	04/10/2015	ARRIVAL FOR SEN HETTRAMP BISMARCK TO FARGO, MINOT AND RETURN	94.25
					STAFF PER DEM	707.60
					STAFF TRANSPORTATION	
					WASHINGTON DC TO BISMARCK, DICKINSON, MANDAN, BISMARCK, MINOT, BISMARCK AND RETURN	
DHET21500340	04/04/2015	MCDONOUGH,ANGELA	04/16/2015	04/16/2015	STAFF TRANSPORTATION	13.00
DHET21500341	04/03/2015	MEYER,SHIRLEY J	04/16/2015	04/16/2015	WASHINGTON DC OFFICE, INTERDEPARTMENTAL TRANSPORTATION	37.10
DHET21500342	04/03/2015	MEYER,SHIRLEY J	04/13/2015	04/13/2015	STAFF TRANSPORTATION	16.14
					DICKINSON TO MEDORA AND RETURN	60.30
DHET21500343	04/23/2015	KEYS,ROSS D	03/19/2015	03/27/2015	STAFF TRANSPORTATION	29.00
DHET21500344	04/24/2015	LONG,JAMIE E	04/14/2015	04/14/2015	BISMARCK OFFICE, INTERDEPARTMENTAL TRANSPORTATION	7.56
					STAFF PER DEM	90.00
DHET21500345	04/03/2015	MCCLOUD,NORMAN F	04/08/2015	04/10/2015	STAFF TRANSPORTATION	98.41
					FARGO TO GRAND FORKS AND RETURN	
					STAFF PER DEM	111.00
					STAFF TRANSPORTATION	
					MINOT TO BISMARCK AND RETURN	
DHET21500348	04/03/2015	HETTRAMP,HEIDI	04/17/2015	04/09/2015	SENATORS TRANSPORTATION	478.20
DHET21500350	04/08/2015	MCDONOUGH,ANGELA	03/30/2015	04/01/2015	WASHINGTON DC TO MINOT, NEW TOWN, BISMARCK AND RETURN	340.85
					STAFF PER DEM	872.25
					STAFF TRANSPORTATION	
					WASHINGTON DC TO GRAND FORKS AND RETURN	

B-1155

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- ▶ This is a complex task, which could either result in intensive coding marathons or years of hand coding
- ▶ I don't want to do either, so I turn to Claude Code

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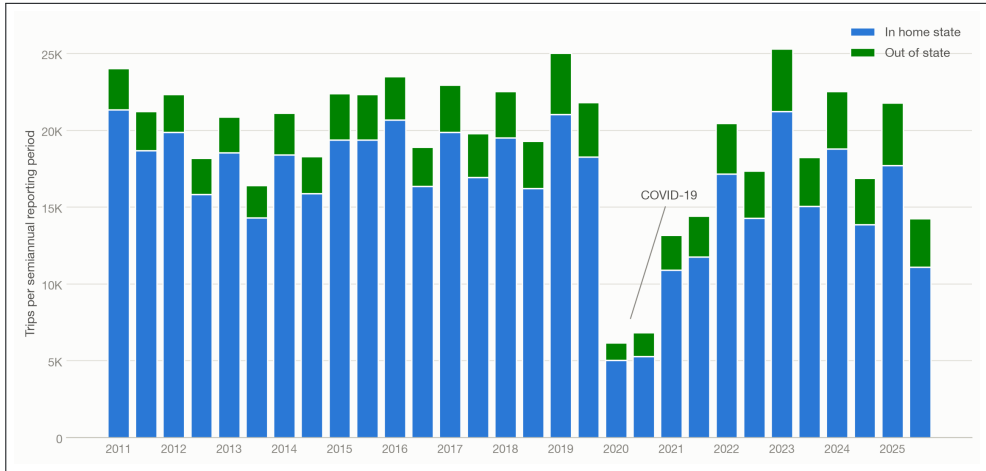
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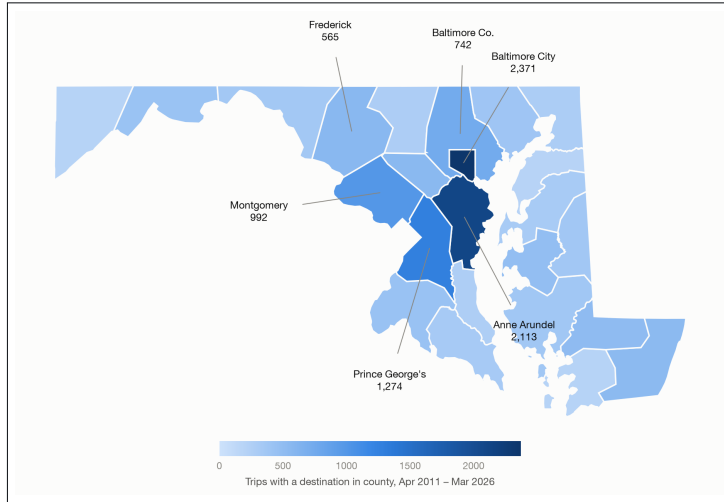
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- ▶ The result: full coverage of all Senator and staff travel from 2011 to 2026, about 1 million total observations
- ▶ Total active working time: 3.2 hours

Application of Agentic Coding in My Research

Distribution of Travel:



Where do Maryland Senators and their Staff Visit?



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- ▶ Scholars from even a few years ago would not have considered a project like this because of the hurdles
- ▶ The “big” ideas of the past are not limited to scholars at the most funded universities

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- ▶ **ALWAYS DOUBLE CHECK!**

- ▶ Introduction to Prompt Engineering

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- ▶ Familiarized yourself with what agentic coding can do

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- ▶ Familiarized yourself with what agentic coding can do
- ▶ Discussed the implications of AI in research

Questions?